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Unit – 4 : Session – 10 : SLO – 1 | B Trees

■ I. Fill in the Blanks

- A B Tree is a SELF-BALANCING (m-way search) tree.
- B Trees are used in DATABASES and FILE SYSTEMS.
- A B Tree of order m has at most m CHILDREN.
- A node can contain at most m-1 keys.
- All leaf nodes appear on the SAME LEVEL.
- B Trees are BALANCED trees.
- Internal nodes have at least $\text{ceil}(m/2)$ children.
- B Trees grow and shrink from the ROOT (top).
- Insertion in B Trees is done at the LEAF LEVEL.
- If a node overflows, it is SPLIT.
- If a node underflows, it may MERGE or REDISTRIBUTE (borrow).
- B Trees eliminate the need for REORGANIZATION (rebalancing of the whole tree) after insertions and deletions.
- B Trees reduce time for DISK (secondary storage) accesses.
- B Trees maintain SORTED data.
- B Trees are generalizations of BINARY SEARCH TREES (2-3 trees).